Xiutian Zhao

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Research Summary

Speech and language processing, interpretability of multimodal foundation models, and controllable generation. My recent work focuses on neuron-level emotion representations in audio-language models, building on earlier work on large language model reasoning and multi-agent systems, with publications in ACL, AAAI, EACL, and EMNLP.

Education

Johns Hopkins University — Baltimore, MD
Ph.D. student, Electrical and Computer Engineering — 2025 - Present
- Advisor: Berrak Sisman
- Affiliation: Center for Language and Speech Processing
  Data Science and AI Institute

University of Edinburgh — Edinburgh, UK
MSc., Speech and Language Processing, with Distinction — 2025
- Advisor: Ivan Titov
- Dissertation: *Culture-Specific Neurons in Vision-Language Models: Identification, Causal Ablations, and Layer-wise Localization*.

Columbia University — New York, NY
M.S., Applied Statistics — 2019

Fordham University — New York, NY
B.A., Interdisciplinary Mathematics & Economics — 2017

Peer-reviewed Publications

[10] Discovering and Causally Validating Emotion-Sensitive Neurons in Large Audio-Language Models
Xiutian Zhao, Björn Schuller, Berrak Sisman
ACL 2026 (Main Conference), **oral presentation**.

[9] Finding Culture-Sensitive Neurons in Vision-Language Models
Xiutian Zhao, Rochelle Choenni, Rohit Saxena, Ivan Titov
EACL 2026 (Main Conference), **oral presentation**.

[8] An Electoral Approach to Diversify LLM-based Multi-Agent Collective Decision-Making
Xiutian Zhao, Ke Wang, Yanghui Li, Wei Peng
EMNLP 2024 (Main Conference), also presented at *ICML 2024 HAS Workshop*.

[7] Watch Every Step! LLM Agent Learning via Iterative Step-level Process Refinement
Weimin Xiong, Yifan Song, **Xiutian Zhao**, Wenhao Wu, Xun Wang, Ke Wang, Cheng Li, Wei Peng, Sujian Li
EMNLP 2024 (Main Conference).

[6] AgentBank: Towards Generalized LLM Agents via Fine-Tuning on 50000+ Interaction Trajectories
Yifan Song, Weimin Xiong, **Xiutian Zhao**, Dawei Zhu, Wenhao Wu, Ke Wang, Cheng Li,

Wei Peng, Sujian Li
EMNLP 2024 (Findings).

[5] Measuring the Inconsistency of Large Language Models in Preferential Ranking
Xiutian Zhao, Ke Wang, Yanghui Li, Wei Peng
ACL 2024 (Workshop on Towards Knowledgeable Language Models), **oral presentation**

[4] Learning from Failure: Improving Meeting Summarization without Good Samples
Ke Wang, **Xiutian Zhao**, Wei Peng
AAAI 2024 (Main Conference).

[3] ORCHID: A Chinese Debate Corpus for Target-Independent Stance Detection and Argumentative Dialogue Summarization
Xiutian Zhao, Ke Wang, Wei Peng
EMNLP 2023 (Main Conference).

[2] PROSE: A Pronoun Omission Solution for Chinese-English Spoken Language Translation
Ke Wang, **Xiutian Zhao**, Yanghui Li, Wei Peng
EMNLP 2023 (Main Conference).

[1] M^3Seg: A Maximum-Minimum Mutual Information Paradigm for Unsupervised Topic Segmentation in ASR Transcripts
Ke Wang, **Xiutian Zhao**, Yanghui Li, Wei Peng
EMNLP 2023 (Main Conference).

Preprints

[4] TRACE-EVC: Text-Guided Relative Affective Control for Zero-Shot Emotional Voice Conversion
Zihan Zhang, Shreeram Suresh Chandra, Zongyang Du, **Xiutian Zhao**, Aurosweta Mahapatra, Hao Zhang, Philipp Koehn, Berrak Sisman
Preprint on arXiv, under review

[3] Neuron-Level Emotion Control in Speech-Generative Large Audio-Language Models
Xiutian Zhao, Ismail Rasim Ulgen, Philipp Koehn, Björn Schuller, Berrak Sisman
Preprint on arXiv, under review

[2] Unwrapping Circularity: Can Transformers Learn Languages with Circular Schemes?
Xiutian Zhao, Rafi Aulia, Siying Chen, Xiulin Yang
Presented at *ACL 2025 Student Research Workshop*

[1] A Survey on Hallucination in Large Vision-Language Models
Hanchao Liu, Wenyuan Xue, Yifei Chen, Dapeng Chen, **Xiutian Zhao**, Ke Wang, Liping Hou, Rongjun Li, Wei Peng
Preprint on arXiv

Research Experience

Research Assistant | Peking University–Huawei Co-Laboratory 2023 - 2024
- Conducted research on LLM-based multi-agent systems and agent learning, resulting in publications at EMNLP 2024.

Research Assistant | Huawei Technologies 2020 - 2024
- Conducted research in machine translation, meeting summarization, and dialogue understanding, resulting in publications at AAAI, EMNLP, and ACL workshops.

Awards and Honors

- **Relocation Grant**, Johns Hopkins University, $2,500 2025
- **The Edinburgh Award**, School of Informatics: Tutoring, University of Edinburgh 2025
- **Psi Chi**, Columbia University 2018

Teaching

Informatics Tutor: *Informatics Tutoring Scheme* 2024 - 2025
A project that supports high school students on Computer Science and Maths, with a priority for schools in disadvantaged areas of Scotland.

Academic Service

Reviewer for: *ACL Rolling Review* 2024, 2025, 2026
IEEE Signal Processing Letters 2026
IEEE Transactions on Affective Computing 2025, 2026
COLM, the 2nd Workshop on Actionable Interpretability 2026
NeurIPS 2025
AAAI 2025
COLM 2025
EACL, Student Research Workshop 2025
ACL, the 11th Workshop on Argument Mining 2024

Student Committee: *MASC-SLL 2026* 2026
Mid-Atlantic Student Colloquium in Speech, Language and Learning.

Organizing Committee: *Linguistic Circle* 2024 - 2025
A flagship seminar series organized by the Linguistics and English Language Department of the University of Edinburgh.

Student Volunteer: *EACL 2026* 2026
EMNLP 2024 2024

Skills

Languages: English, Mandarin Chinese, Japanese

Computing: `Python`, LaTeX, Linux, Slurm / cluster computing